
Rapid mass changes at the core-mantle boundary originating from the deep mantle as seen by the GRACE mission

Charlotte Gagne^{1,3} (Phd candidate)
Isabelle Panet^{1,2}, Marianne Greff¹, Mioara Manda³, Séverine Rosat⁴

¹ Université Paris Cité, Institut de physique du globe de Paris, CNRS, IGN

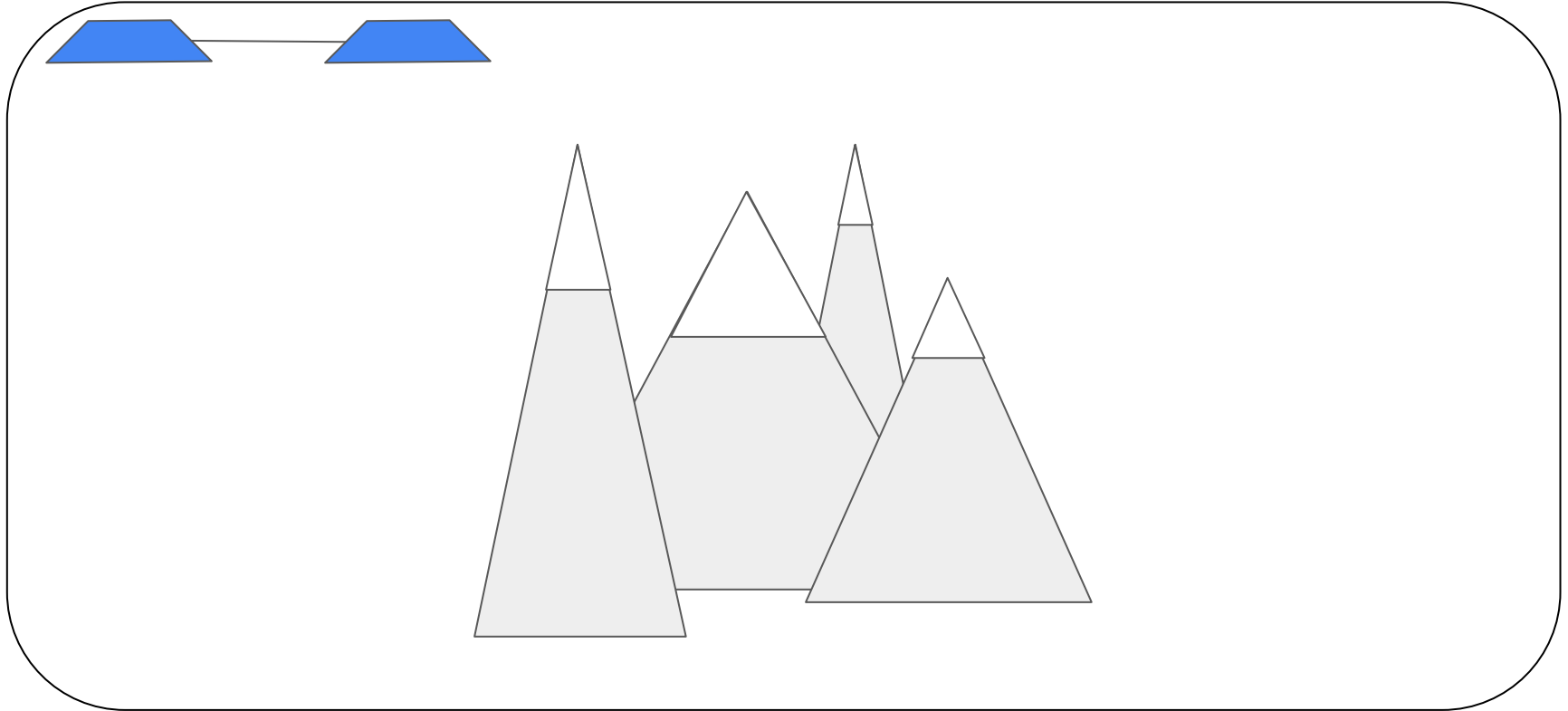
² ENSG Géomatique, IGN, F-77455 Marne-la-Vallée

³ Centre National d'Études Spatiales, Paris

⁴ Université de Strasbourg, CNRS, Institut Terre et Environnement de Strasbourg, UMR 7063

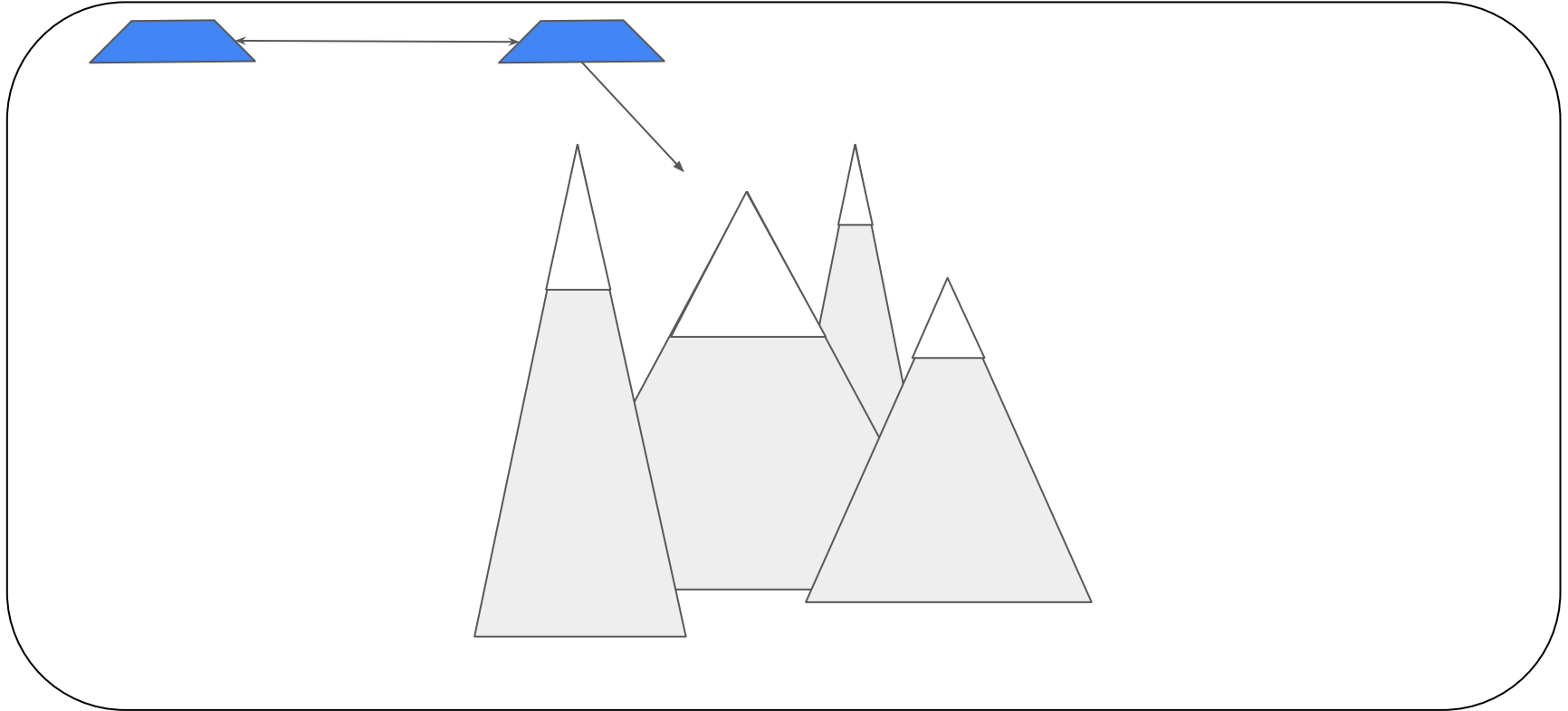


Measurements of the space-time variations of the gravity field from the GRACE & GRACE-FO missions (since 2002)



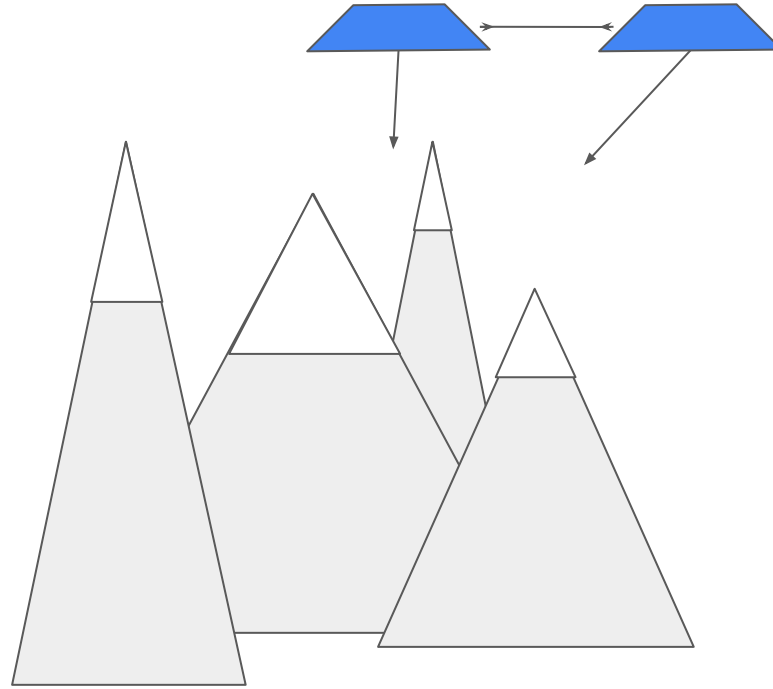


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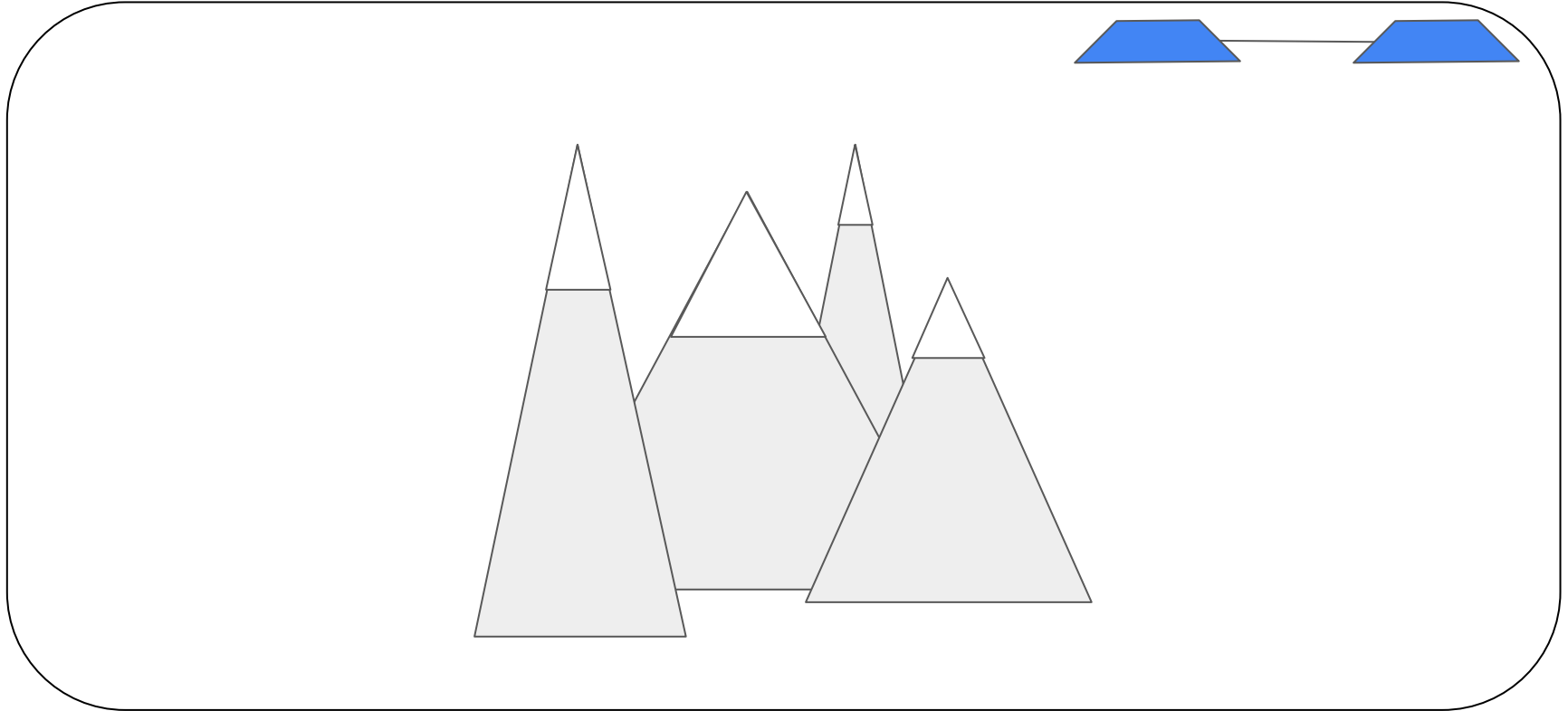


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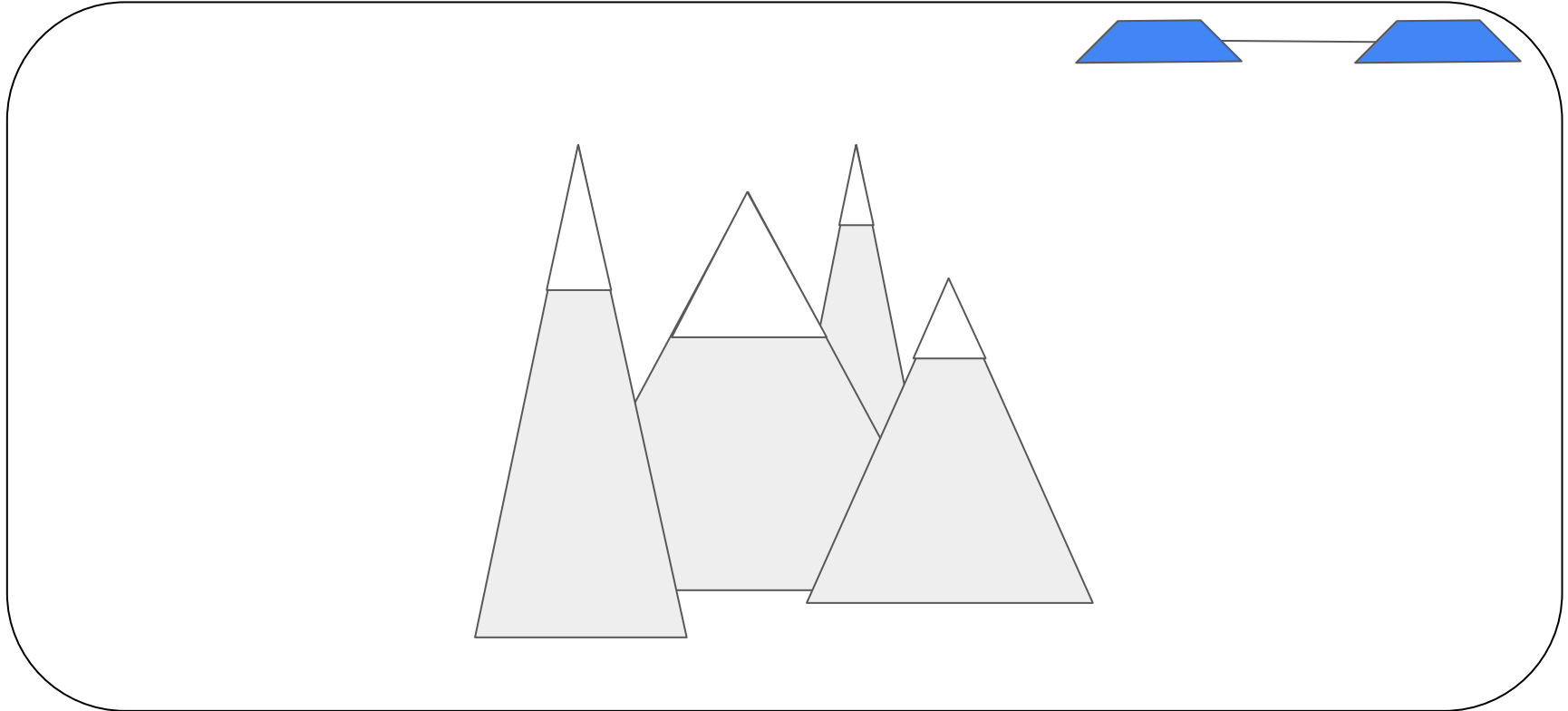


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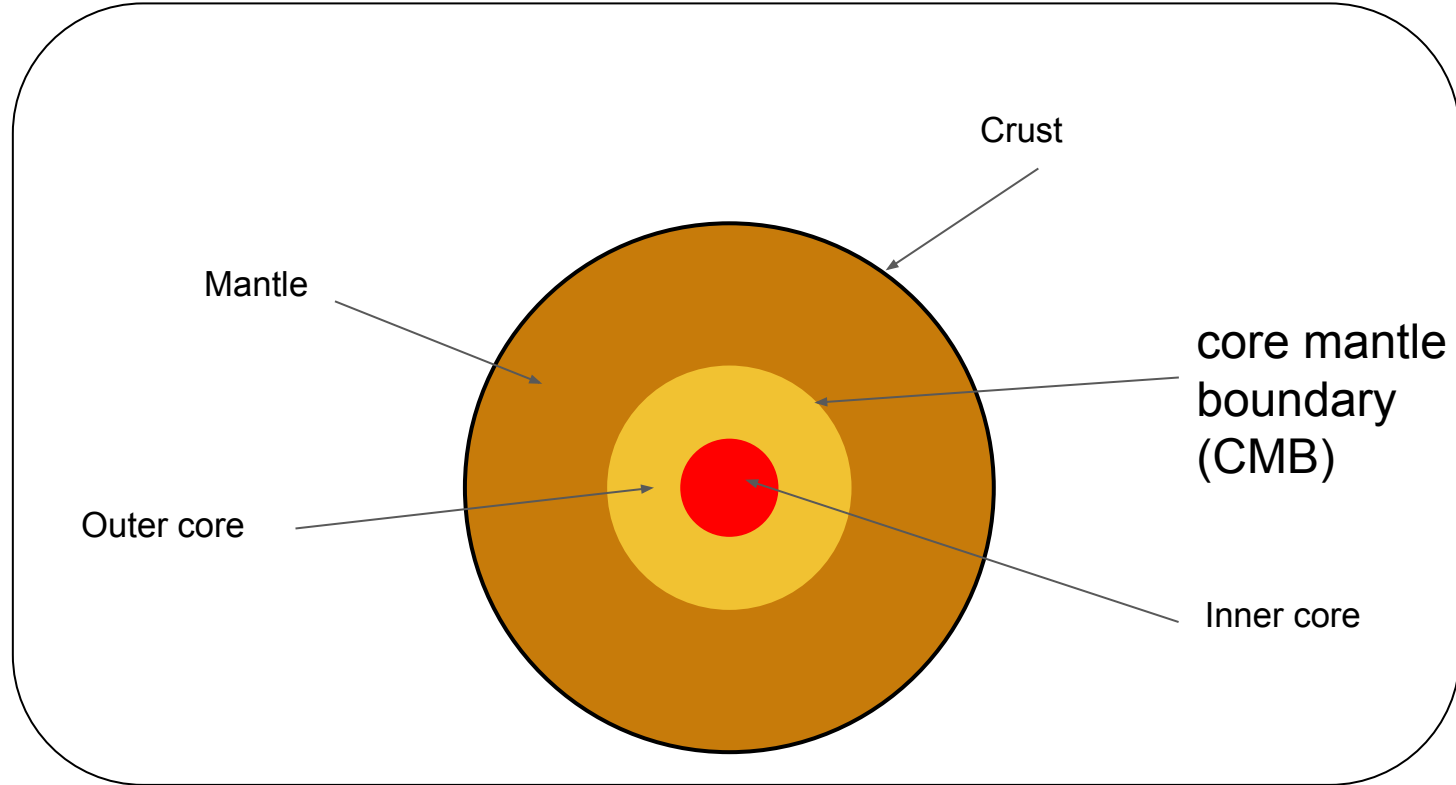


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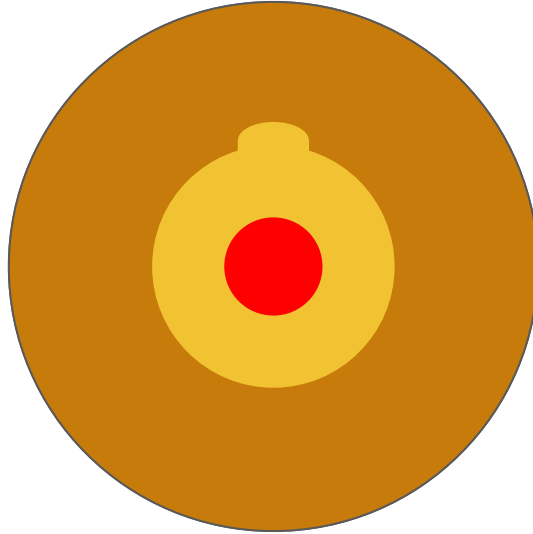


→ **new constraints on deep mass redistributions ?**

Mass redistributions due to variations of the core mantle boundary (CMB) topography:

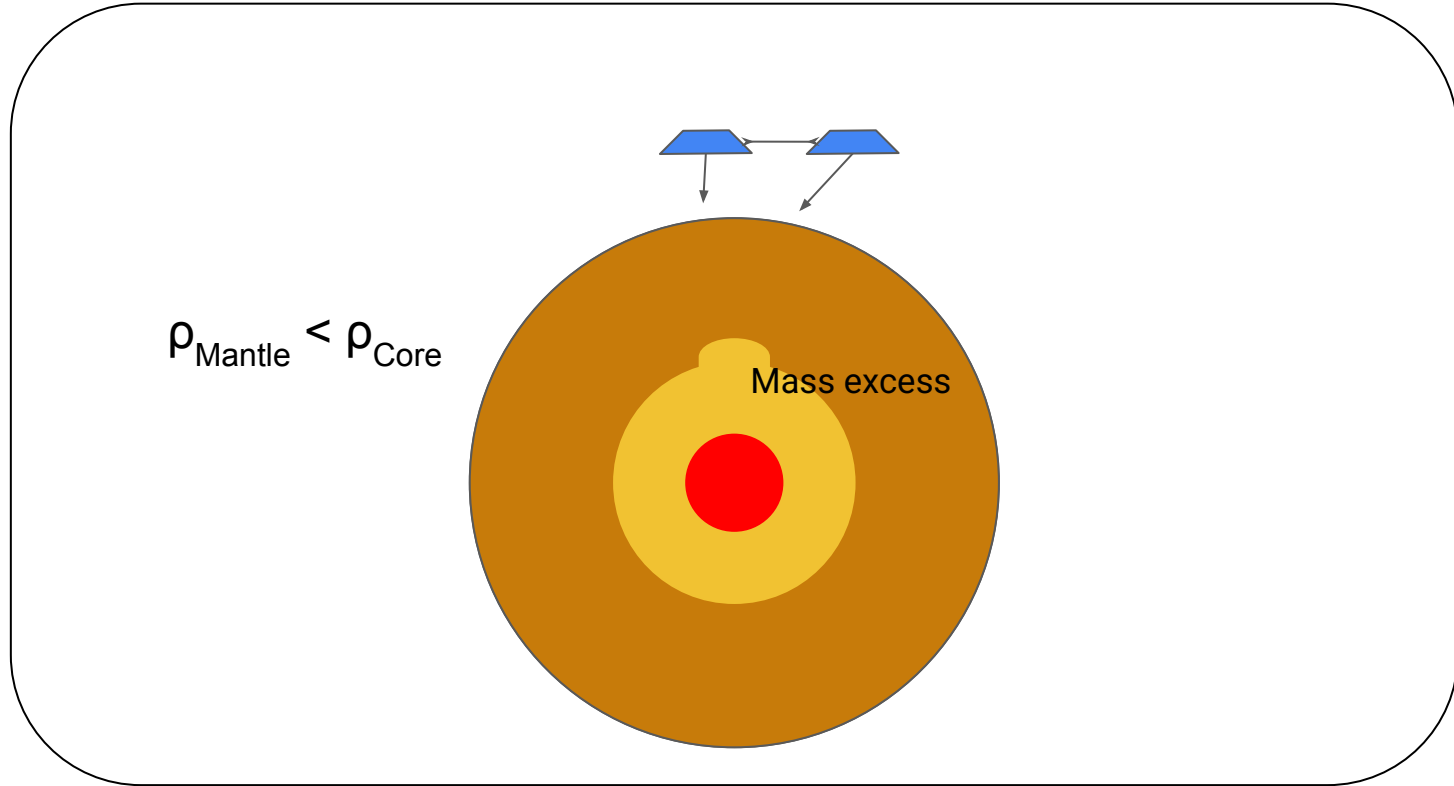


Mass redistributions due to variations of the core mantle boundary (CMB) topography:

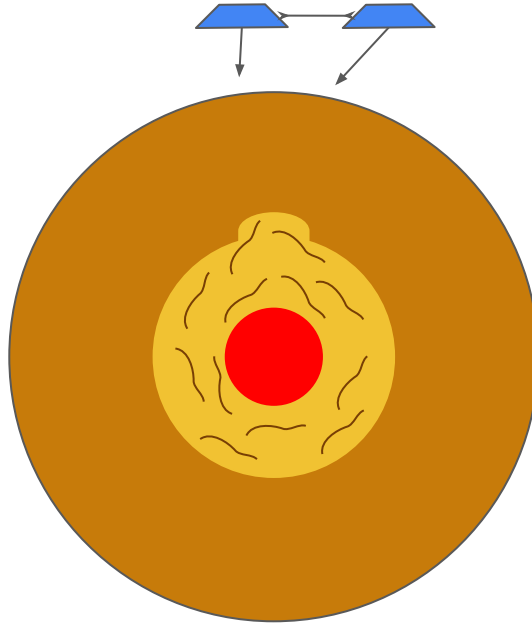


Not to scale

Mass redistributions due to variations of the core mantle boundary (CMB) topography:



Mass redistributions due to variations of the CMB topography: coupling with sudden changes in core flows ?



This could help to better understand sudden changes in the secular variation of the geomagnetic field, the **geomagnetic jerks**.

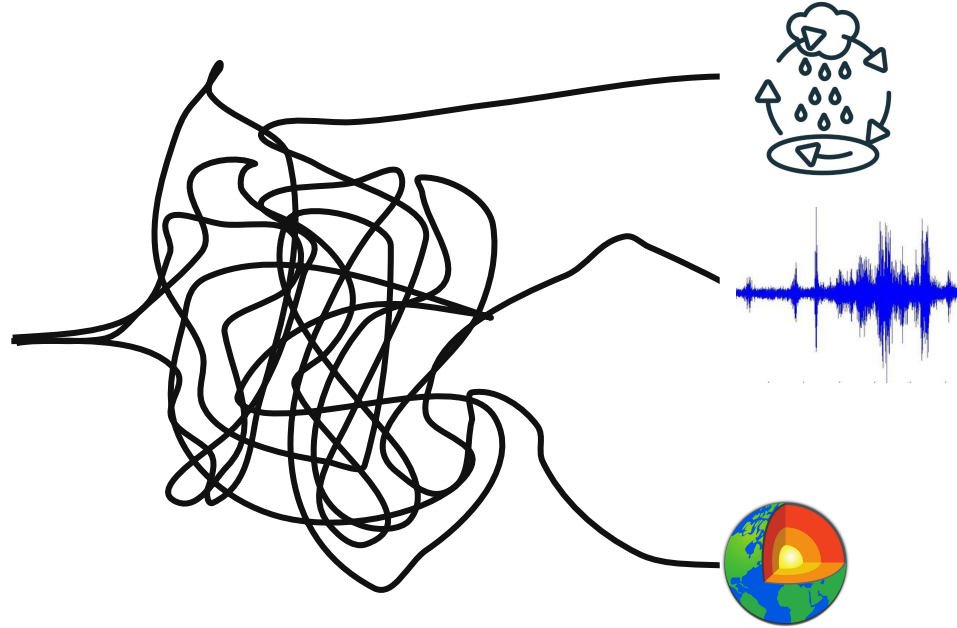
Objectives



Considering the **2007** geomagnetic jerk (*Olsen and Manda* 2008) well observed in the **Atlantic ocean**, we search for a possible corresponding **gravity signal** of mass redistributions at the **CMB** at timescales of **months to a few years**.

→ **Problem:** GRACE signal are mainly due to water

How to separate signals record by GRACE ?

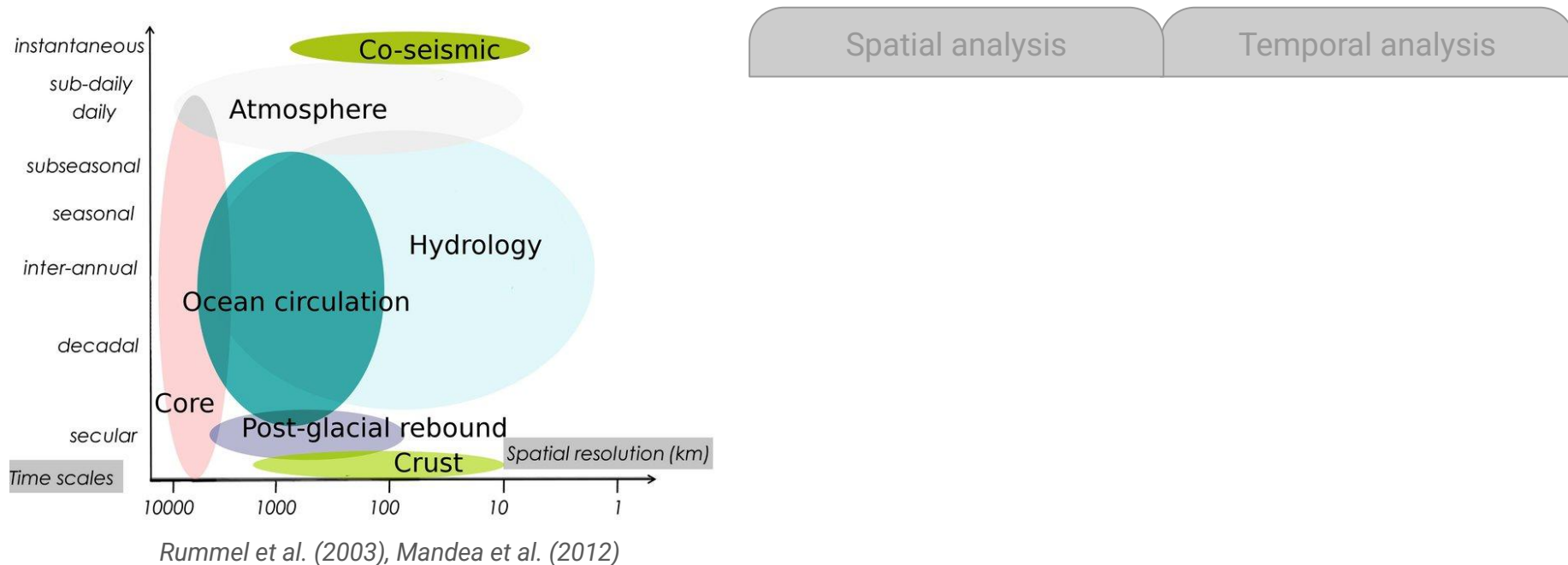


?

Methods

- We analyze 4 sets of GRACE monthly geoid models and 1 geoid model based on SLR data only
- We subtract a **mean**, a **trend**, **annual** and **semi-annual** signals (2003-2015)

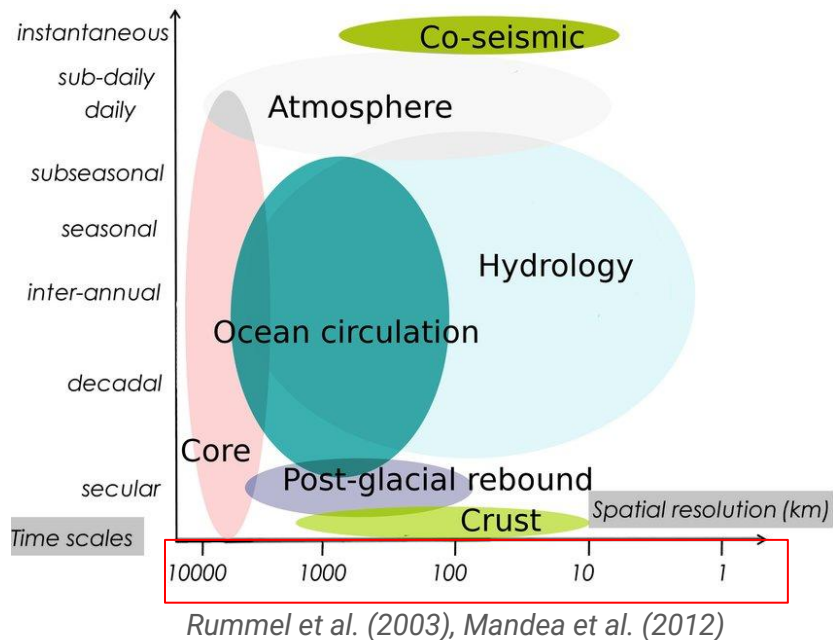
Aim: **separate** signals considering their different **spatial scales**, **orientations** and **temporal scales**.



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Spatial analysis

Temporal analysis

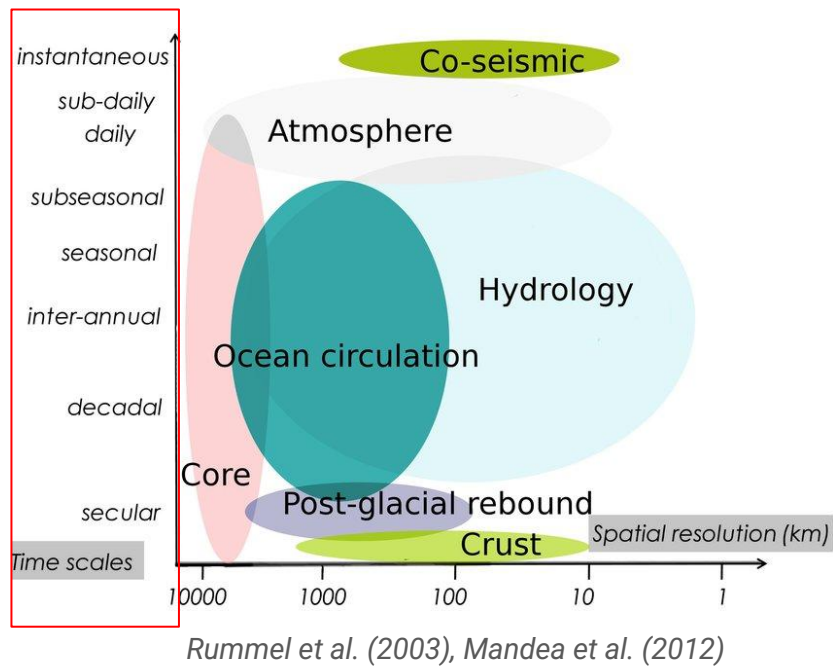
- Filter the geoid at different spatial scales using a **wavelet analysis** to focus on **large scales** which are those of **deep** signals
- For each scale, the geometry of the signals is described by calculating the **gravity gradients** in the local spherical frame (no degree 0 or 1 terms)
 - **Rotations of the spherical frame** to align with the orientation of the signals

(Panet et al. 2018 ; Panet 2019)

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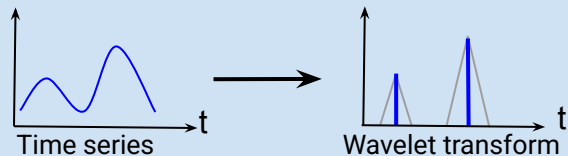
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Spatial analysis

Temporal analysis

- Time series of gravity gradients at different spatial scales > transient signals via wavelet transform
- **Bump** in the time series ↔ **peak** in the wavelet-transform coefficients

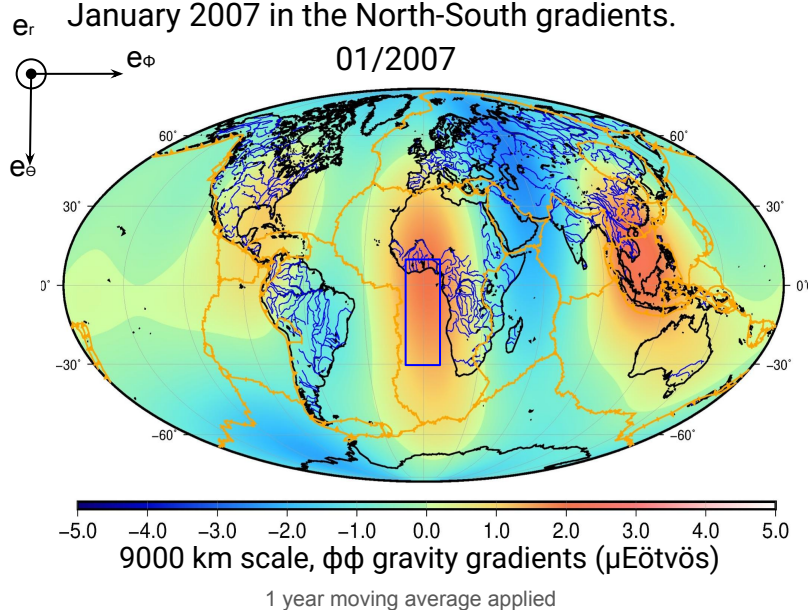


- Search for peaks over the period **September 2006 - April 2007, at monthly to inter-annual timescales**

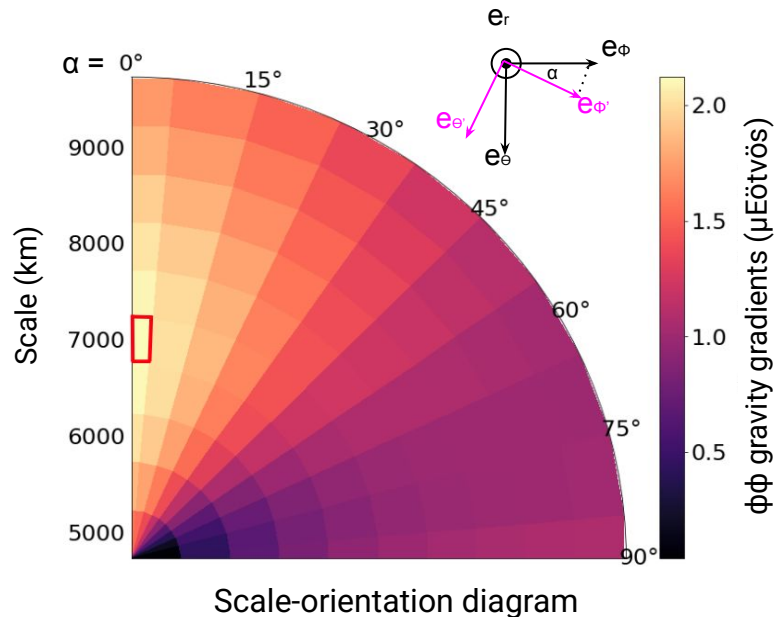
(Panet et al. 2018 ; Panet 2019)

Detection of an anomalous signal in the GRACE data

In the **time series** of large-scale gravity gradients obtained at each point of the globe, we observe an intense signal in January 2007 in the North-South gradients.

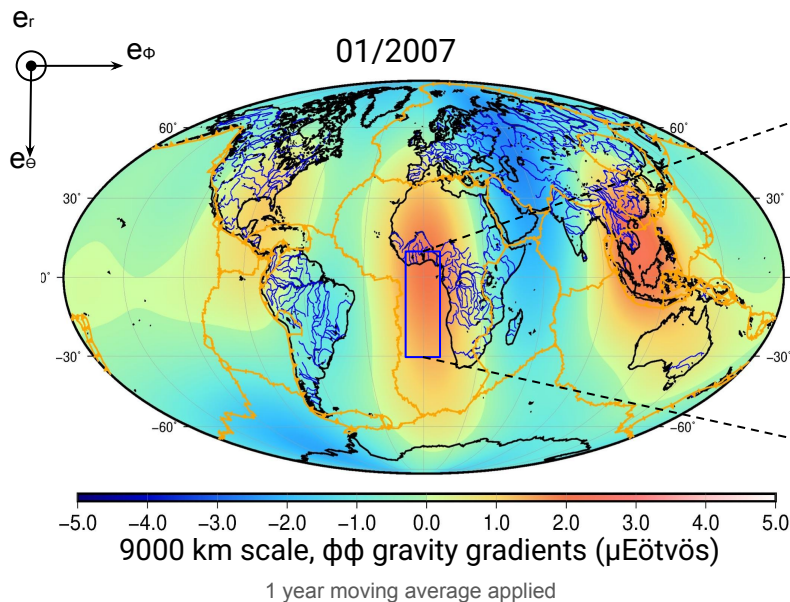


GRACE geoid model: GRGS04

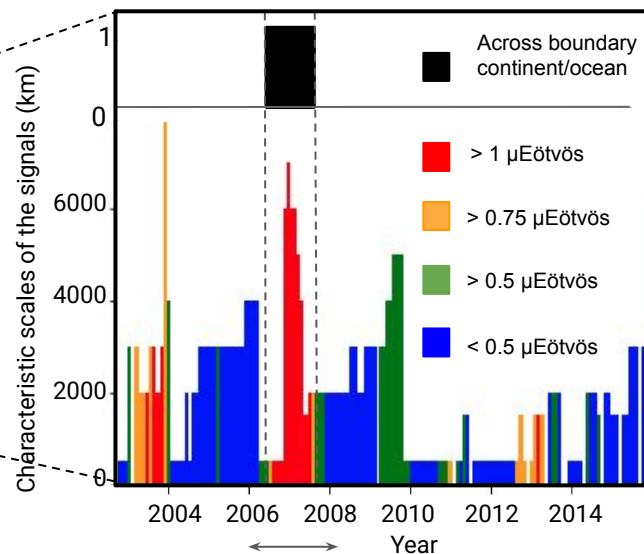


→ Anomalous 7000 km scale, North-South oriented signal **across** the boundary between the **Atlantic ocean** and the **African continent** in January **2007**. Same **timing** and **location** as the 2007 geomagnetic jerk.

Detection of an anomalous signal in the GRACE data



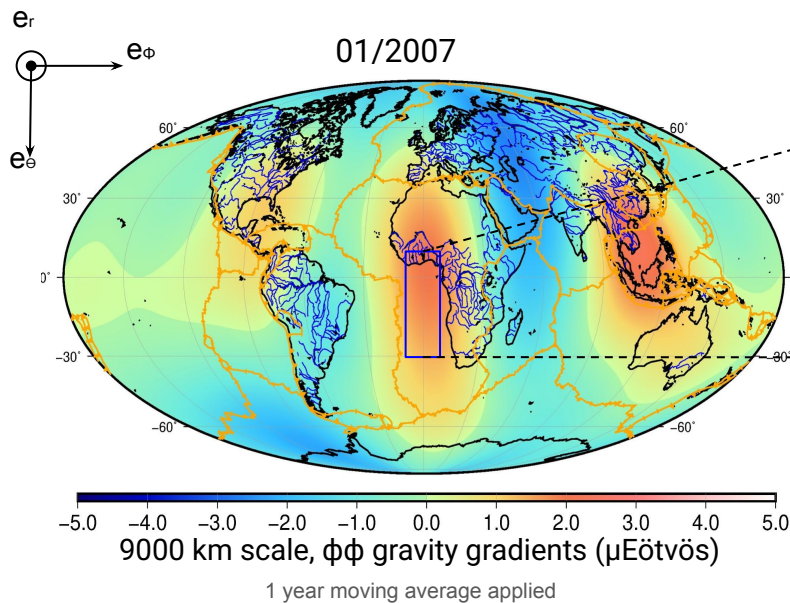
GRACE geoid model: GRGS04



Characteristic spatial scale and intensity of the signals in the studied region, over the 2003 - 2015 period

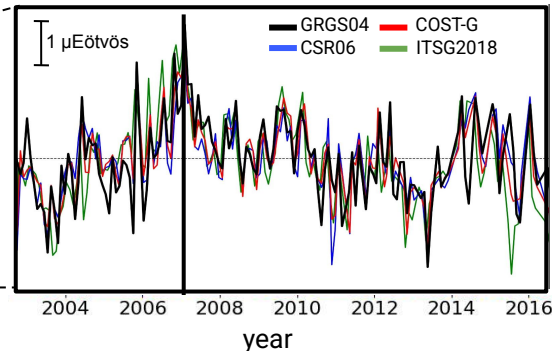
→ In the considered region: **highest** intensity, nearly **largest** spatial scale over 2003-2015

Detection of an anomalous signal in the GRACE data



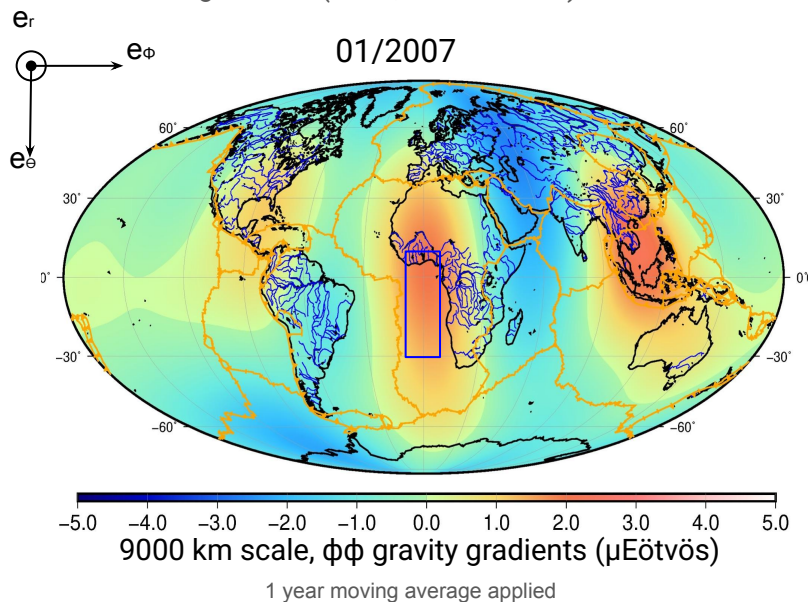
GRACE geoid model: GRGS04

Averaged time series in the Atlantic region for 4 GRACE solutions



Detection of an anomalous signal in the GRACE data

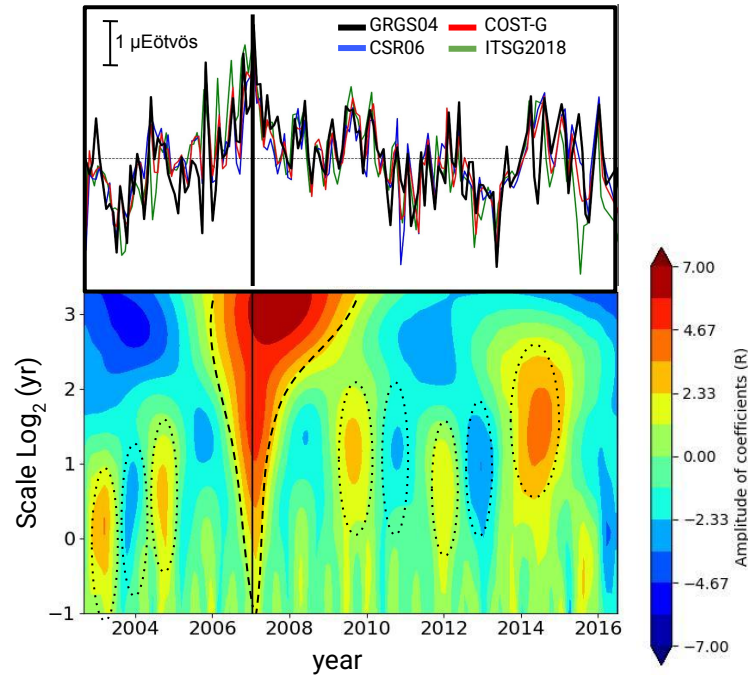
Gaugne et al. (2024, under review)



GRACE geoid model: GRGS04

- **Intense discontinuity** in the timeseries
- Differs from a **2-3 year cycle** signal observed over the **continent**

Averaged time series in the Atlantic region for 4 GRACE solutions

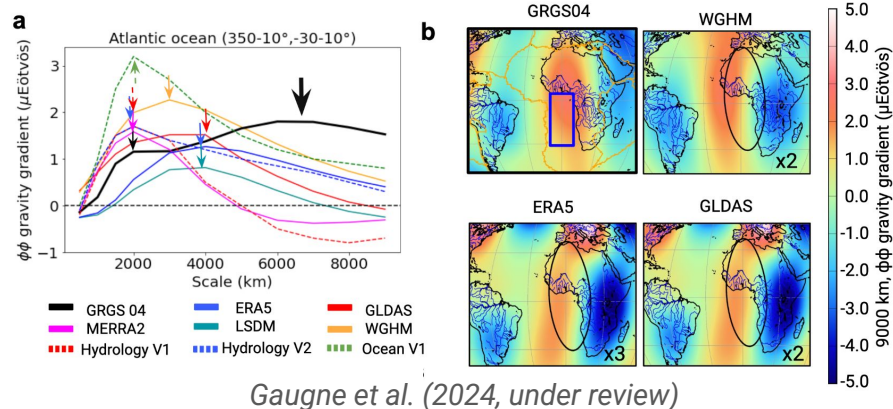


CWT of the 9000 km, $\phi\phi$ gravity gradient time series
GRACE geoid model: GRGS04

A source originating ...

... in the fluid envelope ?

For continental hydrology and oceanic sources (modelled or reconstructed from GRACE), their combinations, and tests of coalescence of continental sources, the **characteristic scale** and **location** differ from those of the 2007 GRACE anomaly.

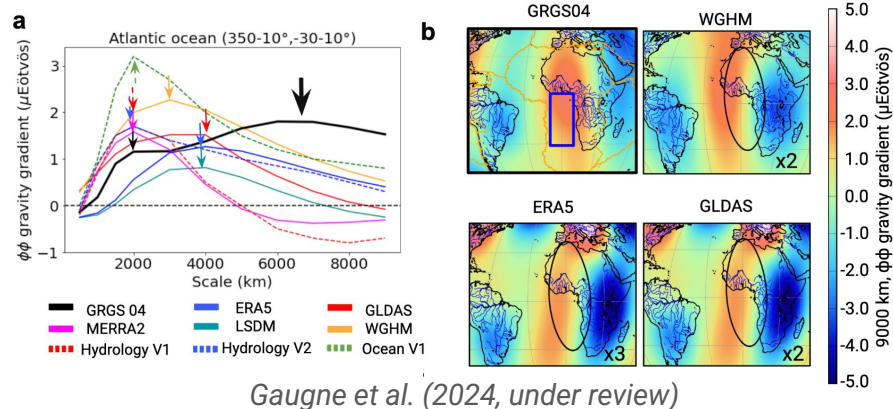


→ The 2007 Atlantic signal is **not well explained** by surface **water** sources.

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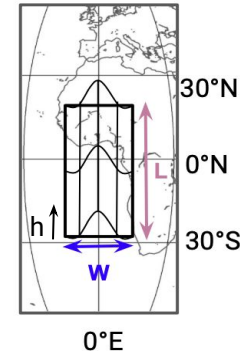
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... Core-Mantle Boundary deflection due to core flows?

Which **amplitude** and **spatial extent** (L x W) of **CMB deformation** could explain the gravity signal ?



L = 5000 km
W = 2000 km
h = 60 cm

→ The gravity-based CMB topography changes presented here are **too large** to be explained by the **dynamic pressure of the core flows**

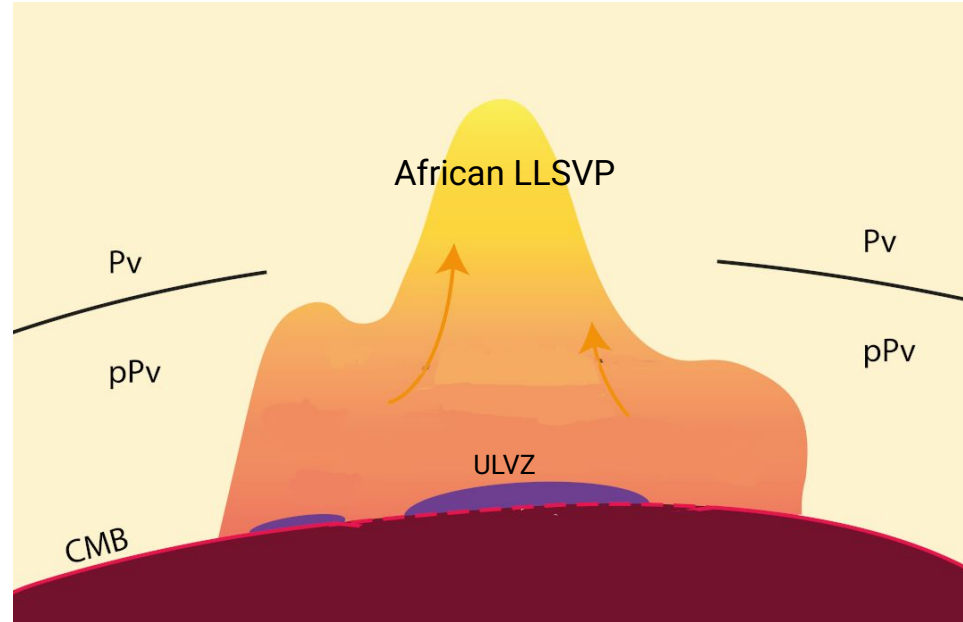
A source originating ...

What **other mechanism** could cause **mass redistribution** at the **CMB** on timescales of months to few years ?

Mass redistribution related to a mineral phase transition

Source in the **mantle above** the CMB:

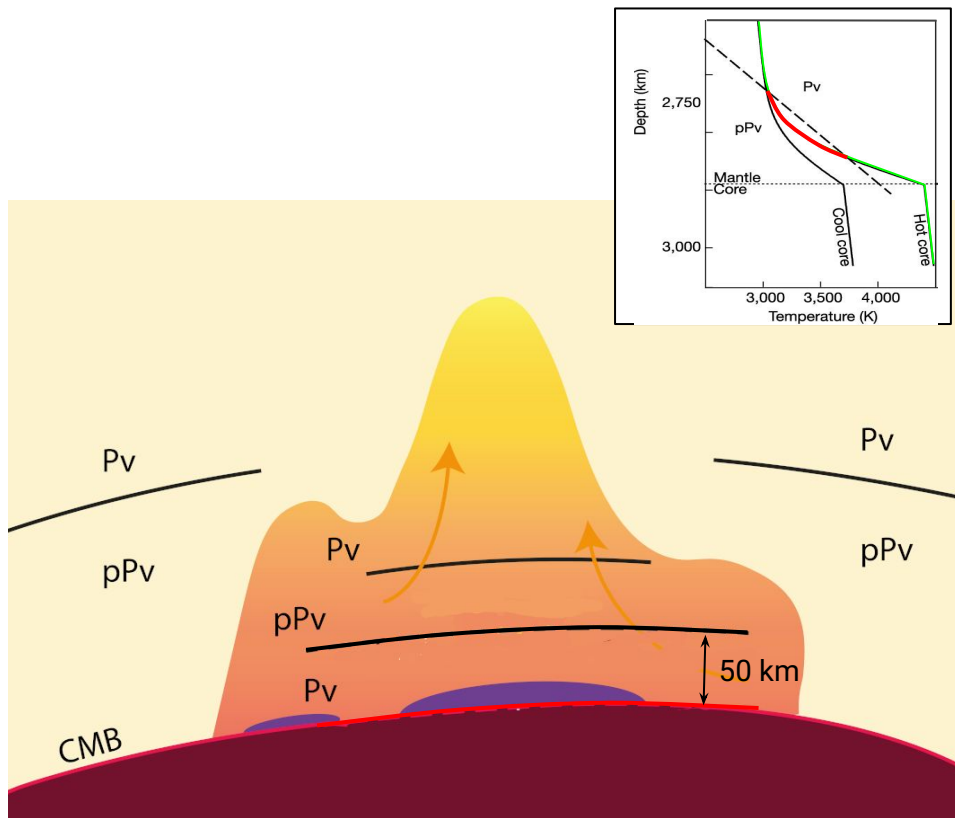
- Perovskite (Pv) to post-perovskite (pPv) phase change: **fast** (Langrand et al., 2019)



Mass redistribution related to a mineral phase transition

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- African LLSVP: **Pv-pPv-Pv phase transition**, deeper in hotter regions (*Wyssession and Solomatov, 2005*)



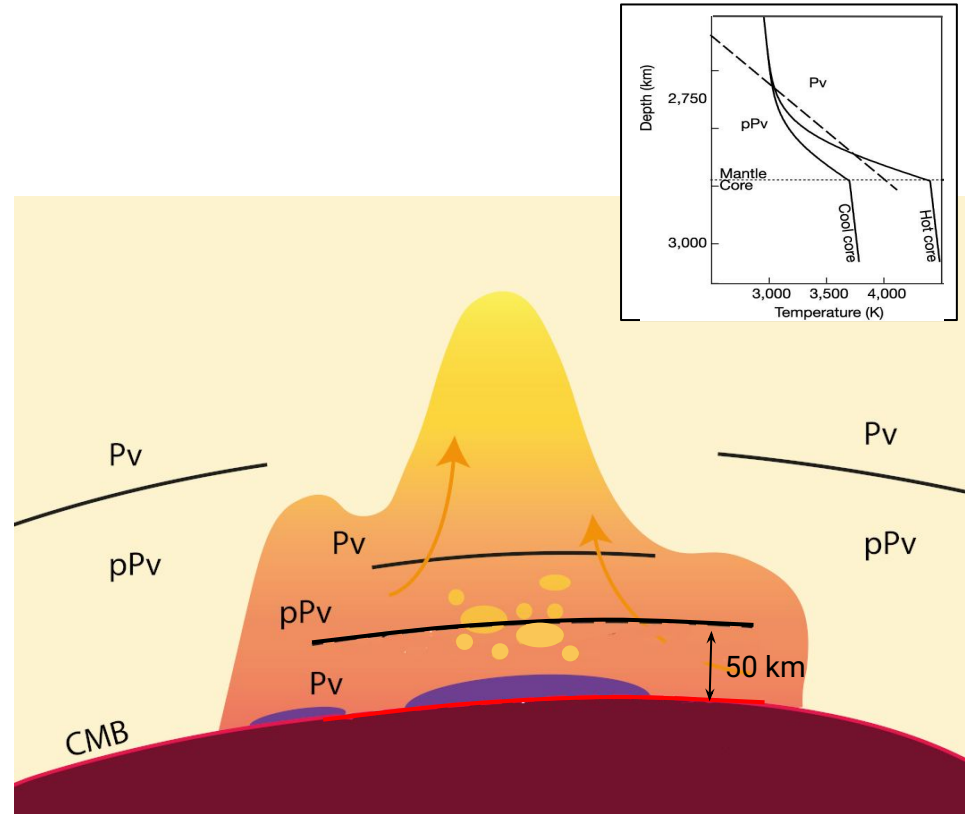
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Mechanism:

- Vertically **moving thermal heterogeneities** entrained in the LLSVP upwellings



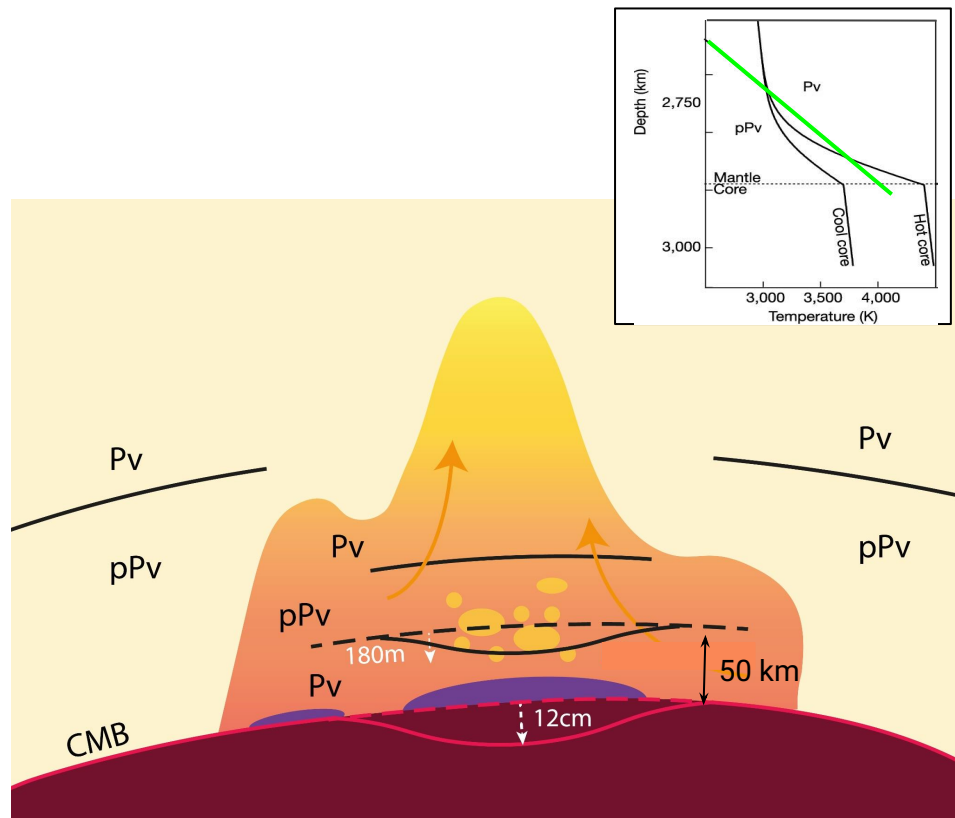
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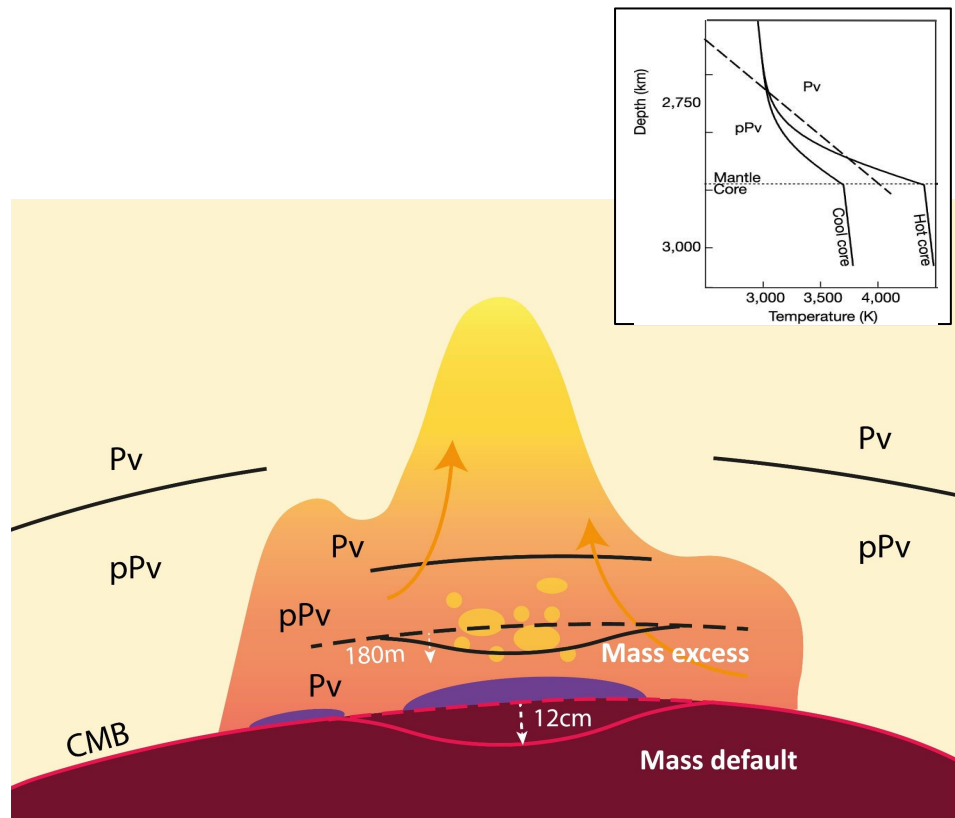
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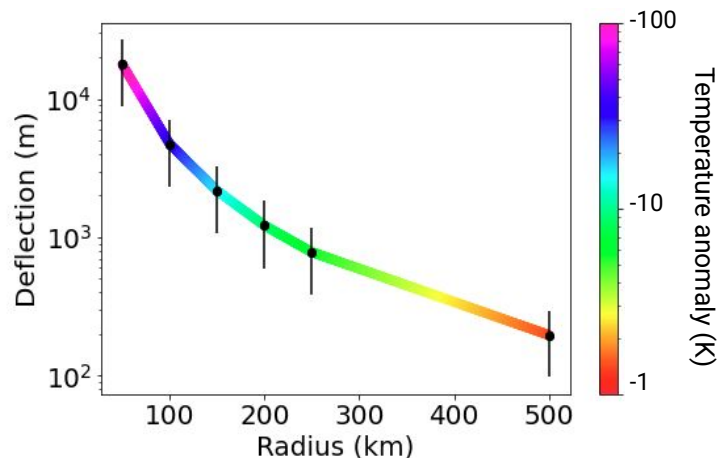
- Vertically **moving thermal heterogeneities** entrained in the LLSVP upwellings
- Pv/pPv large positive Clapeyron slope $\rightarrow$ large **deflection** of the interface
- Pv/pPv density contrast (100 kg/m^3) $\rightarrow$ **mass redistributions**



Mass redistribution related to a mineral phase transition

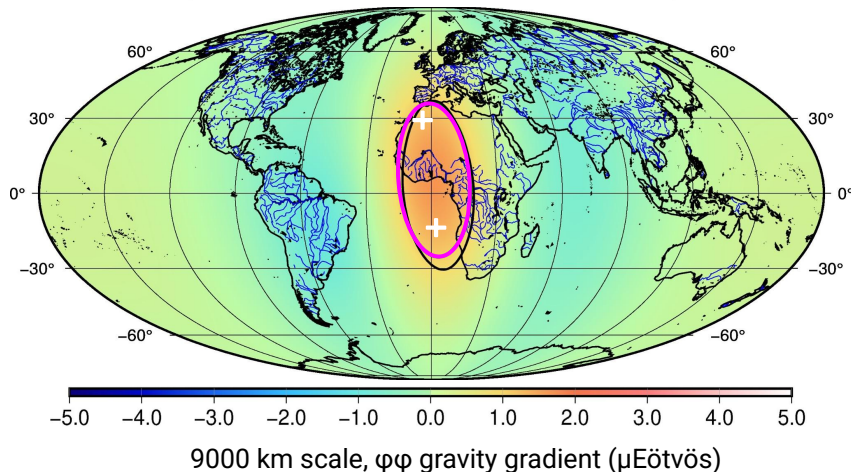
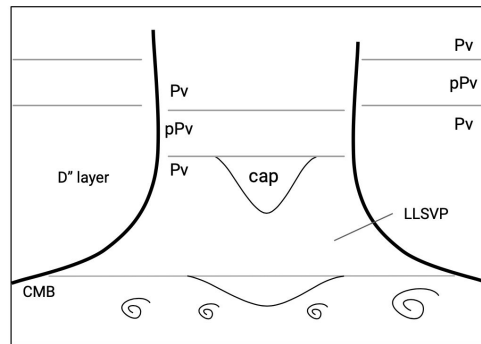
Mass anomalies represented as **2 gaussian caps** at the **base of plumes**. To compare with the GRACE-observed signal, we **add** estimated **hydrological** signals.

Variables radius of caps: 50, 100, 150, 200, 250 & 500 km



Observed signal **well reproduced** with e.g.:

- $R = 500$ km and $\Delta T = -1$ K
- Pv/pPv interface deflection of **180 m**
- **12 cm** of topography at **CMB** and 0.4 mm of geoid



Conclusions

- **Robust** detection of an anomalous signal in four GRACE and one SLR geoid models in **2007** under the **Atlantic ocean** with a timescale of **months to 2-3 years**.
- The **hydrological** models are **not able to explain** the 2007 signal (different spatial scales / different localizations). Same conclusion for oceanic models, for the combination of hydrology + ocean models, and for GRACE-reconstructed hydrological loads. *Confirmed with tests of coalescence of synthetic sources and by investigation of precipitation data over Africa.*
- A deeper origin:
 - At the CMB:
 - Amplitude of **CMB** deformation **60 cm** over a **continental-size** area (1 source) : too large?
 - In the D" layer associated with the **Pv/pPv phase transition**:
 - Amplitude of deformation at **CMB 12 cm** in a **1000 km** wide area due to thermal anomalies of a **few Kelvin** at the base of plumes

What next: Coupling of these mantle-induced **CMB perturbations** with **core flows** and fast **geomagnetic field variations** ?

Thank you for your attention !

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